



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
28-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

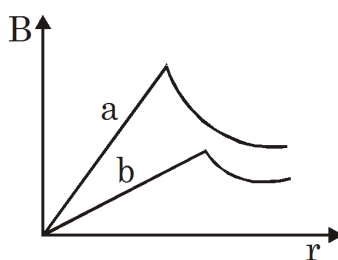
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

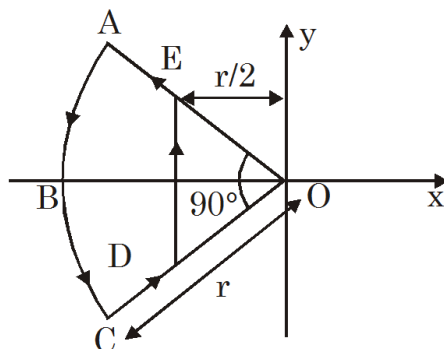
1. Graph shows the variation of magnetic induction due to long straight wires a and b carrying currents uniformly distributed through their cross-sections.



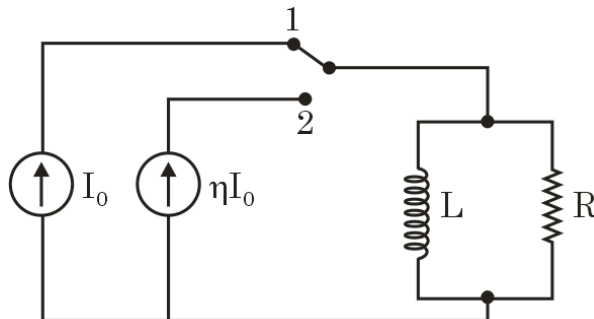
- (A) Radius of wire a is greater than radius of wire b.
 (B) Radius of wire b is greater than radius of wire a.
 (C) Current density in wire a is less than in wire b.
 (D) Current in wire a is less than current in wire b.

Space for Rough Work

2. A wire loop ABCDEA carrying a current i is placed in X-Y plane as shown. The external magnetic field $\vec{B} = B_x \hat{i} + B_z \hat{k}$ is applied in the region. Then



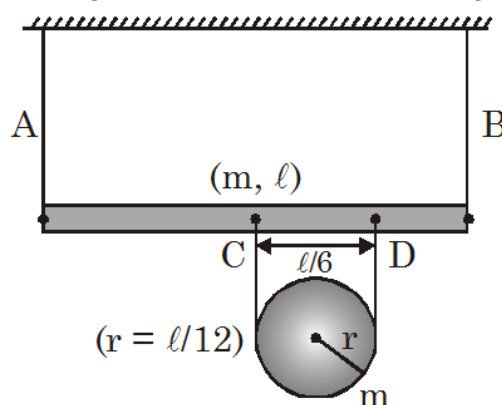
- (A) torque acting on the loop is $\frac{ir^2}{4}(\pi - 1)B_x$.
- (B) torque acting on the loop is $\frac{ir^2}{4}\left(\pi - \frac{1}{2}\right)B_z$.
- (C) net force acting on the loop is $12\pi r\left(\pi - \frac{1}{2}\right)B_x$.
- (D) net force acting on the loop is zero.
3. A parallel combination of an inductance L and a resistance R can be connected to either of the two ideal current sources of current I_0 and ηI_0 through a switch as shown. Initially the switch is in position 1 for a long time. Now the switch is thrown to position 2 until a steady state is established.



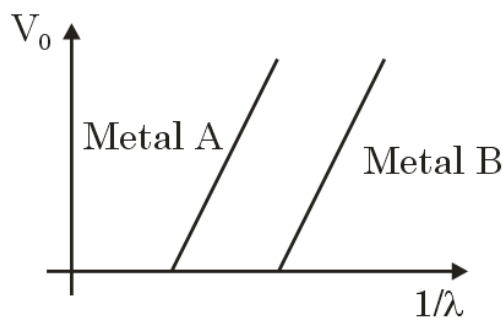
- (A) The charge that will pass through the resistance is $(\eta - 1) \frac{LI_0}{R}$.
- (B) The charge that will pass through the resistance is $\frac{\eta LI_0}{R}$.
- (C) The heat dissipated in the resistor will be $\frac{(\eta - 1)^2 LI_0^2}{2}$.
- (D) The heat dissipated in the resistor will be $\frac{(\eta^2 - 1) LI_0^2}{2}$.

Space for Rough Work

4. A uniform rod of mass ' m ' and length ' ℓ ' is held horizontally by two vertical strings of negligible mass and disc of mass ' m ' is also held by two strings with the rod as shown in the figure. (radius of disc $r = \ell/12$). Then,



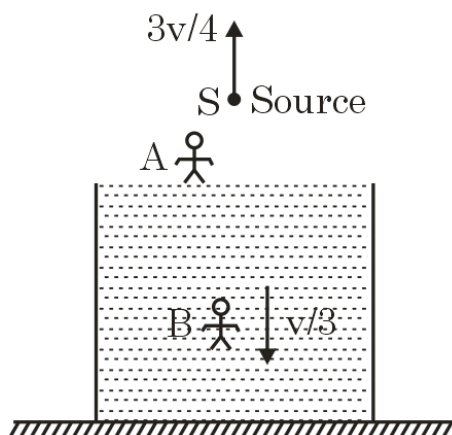
- (A) The tension in the string 'A' immediately after the string B and D is cut is $mg/4$.
 (B) The tension in the string 'A' immediately after the string B and D is cut is $mg/2$.
 (C) The tension in the string 'A' immediately after the string B and C is cut is $mg/4$.
 (D) The tension in the string 'A' immediately after the string B and C is cut is $mg/2$.
5. In an experiment on photoelectric effect, graph of stopping potential V_0 against reciprocal of wavelength λ of the incident light for cathodes of two different metals A and B are as shown. Choose the correct option(s).



- (A) Work function of metal A is smaller than that of metal B.
 (B) Threshold frequency of A is greater than that of metal B.
 (C) When light of certain wavelength is incident on both the cathodes maximum kinetic energy of photoelectrons emitted from A will be greater than that of photoelectrons emitted from B.
 (D) Metal A and B have the same threshold frequency.

Space for Rough Work

6. A liquid is filled in a large container as shown. A source 'S' is moving with constant velocity $3v/4$ in upward direction. A man 'A' is stationary on the surface of water such that his ear is in the air. Another man 'B' is moving downward inside the water with constant velocity $v/3$. Source emits sound of frequency 'f'. Speed of sound in the air is v and in the liquid is $5v$. Then:



- (A) Wavelength received by man A is $\frac{7v}{4f}$. (B) Wavelength received by man B is $\frac{35v}{4f}$
 (C) Frequency received by man A is $\frac{2f}{7}$ (D) Frequency received by man B is $\frac{8f}{15}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

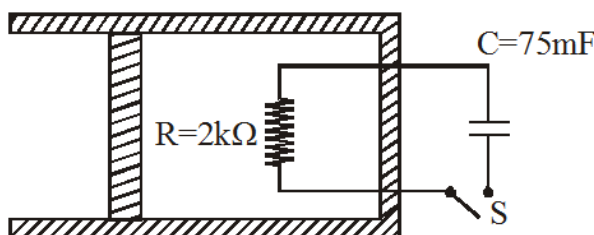
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A cylinder fitted with a piston which can slide without friction contains one mole of an ideal gas. The walls of the cylinder and piston are adiabatic. The cylinder contains a resistor of resistance $R = 2 \text{ k}\Omega$ which is connected to a capacitor of capacity $C = 75 \text{ mF}$. Initially potential difference across capacitor is $\left(\frac{640}{3}\right) \text{ V}$ and switch is open. When switch is closed for $(2.5 \ln 4) \text{ min}$, the gas expands isobarically and its temperature increases by 72 K . Heat loss through the wires is negligible. ($R = 8.3 \text{ J mole}^{-1} \text{ K}^{-1}$)

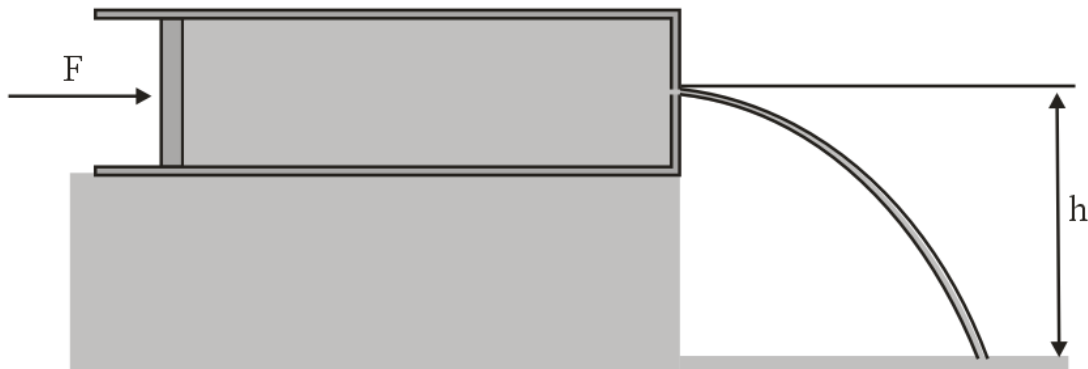


7. Work done by the gas is approximately :
 (A) 0.2 kJ (B) 0.4 kJ (C) 0.6 kJ (D) 0.8 kJ
8. Increment in internal energy of gas is :
 (A) 1 kJ (B) 2 kJ (C) 3 kJ (D) 4 kJ

Space for Rough Work

Paragraph for Questions 9 and 10

An ideal liquid of density ρ is filled in a horizontally fixed syringe fitted with piston. There is no friction between the piston and the inner surface of the syringe. Cross-section area of the syringe is A . At one end of the syringe, an orifice of negligible cross-section area is made. When the piston is pushed into the syringe, the liquid comes out of the orifice following parabolic path and falls on the ground.



9. With what velocity the liquid comes out of the orifice?

(A) $\sqrt{\frac{F}{\rho A}}$ (B) $\sqrt{\frac{2F}{\rho A}}$ (C) $\sqrt{\frac{F + 2\rho ghA}{\rho A}}$ (D) $\sqrt{\frac{3F + 2\rho ghA}{\rho A}}$

10. With what speed the liquid strikes the ground? Neglect the air drag.

(A) $\sqrt{\frac{F + \rho ghA}{\rho A}}$ (B) $\sqrt{\frac{F + 2\rho ghA}{\rho A}}$ (C) $\sqrt{\frac{2F + \rho ghA}{\rho A}}$ (D) $\sqrt{\frac{2(F + \rho ghA)}{\rho A}}$

Space for Rough Work

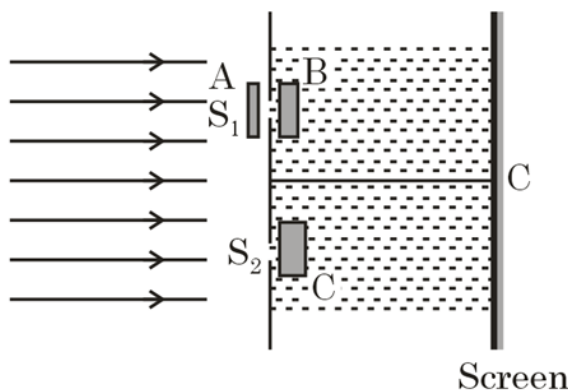
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

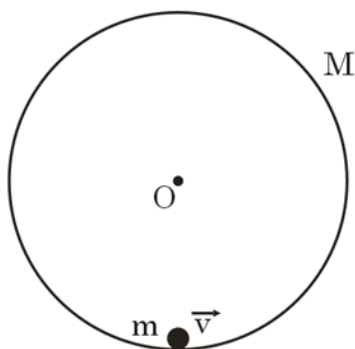
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Three slabs A, B and C are placed in front of slits in YDSE as shown in figure. Refractive index of slabs A, B and C are $\frac{4}{3}$, $\frac{5}{3}$ and $\frac{8}{3}$ respectively and thickness of A, B and C are 2 mm, 1 mm and 1.5 mm respectively. A liquid of refractive index $\frac{2\mu}{3}$ is filled between slits and screen. A monochromatic light of wavelength $5000 \text{ \AA}$ is incident on the slits. If the central bright fringe is obtained at C then find the value of ' μ '.

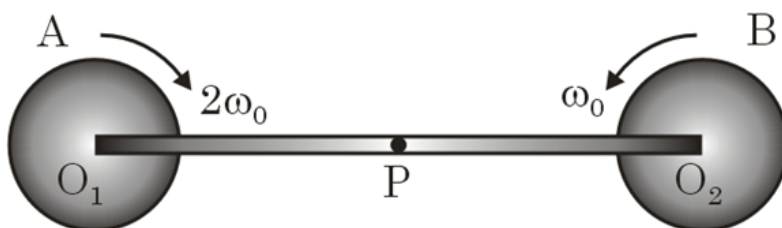


2. A ring of mass $M = 90 \text{ gram}$ and radius $R = 3 \text{ meter}$ is kept on a frictionless horizontal surface such that its plane is parallel to horizontal plane. A particle of mass $m = 10 \text{ gram}$ is placed in contact with the inner surface of ring as shown in figure. An initial velocity $v = \sqrt{2} \text{ m/s}$ is given to the particle along the tangent of the ring. Find the magnitudes of the force of interaction in milli-newton between them after 1 sec from the start.



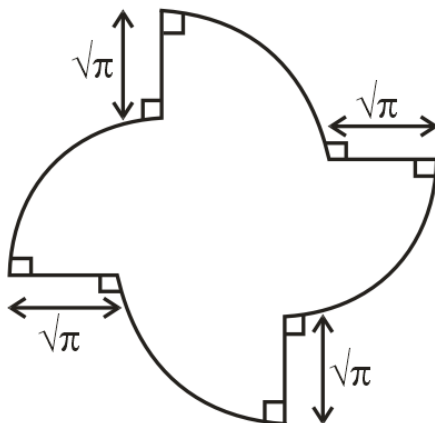
Space for Rough Work

3. Nuclei X and Y convert into a stable nucleus Z. At $t = 0$ the number of nuclei of X is 8 times that of Y. Half life of X is 1 hour and half life of Y is 2 hour. Nucleus X is converted into Z by emitting 4α and 8β particles while nucleus Y is converted into Z by emitting 2α and 4β particles. Find the time (in hour) at which rate of disintegration of X and Y are equal.
4. A rod of mass 'M' and length $6R$ is pivoted at its centre (point P) to rotate in vertical plane as shown. Two discs of same radius 'R' are joined at the ends O_1 and O_2 of the rod. Mass of the discs A and B are 'M' and '2M' respectively. Initial angular velocities of discs are shown in figure. The system is released from rest from horizontal position. The angular velocity of the rod when it is vertical is $\sqrt{\frac{g}{\lambda R}}$. Find the value of λ .

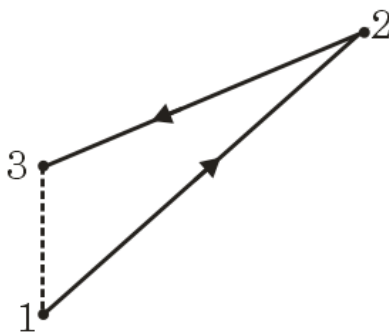


Space for Rough Work

5. As shown in the figure, the radius of each of quarter circular arc is 2m. Find the magnetic moment of the given loop for the current in the loop. Given $i = 0.4$ Amp

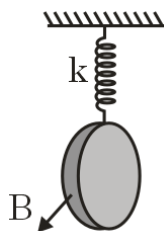


6. In an old record found in a laboratory, a process $1 \rightarrow 2 \rightarrow 3$ was shown. Over the time, ink faded and it became impossible to see the pressure and volume axes. However, descriptions given there reveal that the states 1 and 3 lie on an isochore corresponding to a volume V , amount of heat supplied during the whole process $1 \rightarrow 2 \rightarrow 3$ is zero and the gas involved is one mole of helium. The volume occupied by the gas in the state 2 is nV . Find n .



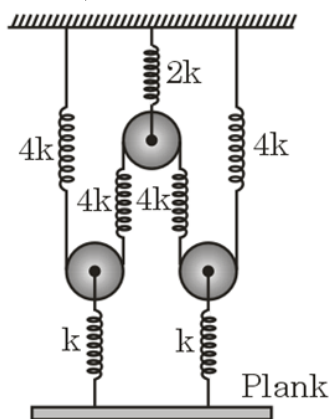
Space for Rough Work

7. A conducting disc of mass m and volume V is suspended with the help of a light spring of force constant k from a fixed support. Thickness of the disc is much smaller than its radius. A uniform magnetic field of induction B parallel to the plane of the disc is established and the disc is pulled down slightly and released. Find period (in sec) of the small amplitude oscillations of the disc. (Given : $\epsilon_0 V B^2 = 6\text{kg}$, $m = 2\text{kg}$, $k = 32\text{ N/m}$)



8. Seven ideal massless springs and three smooth massless pulleys are connected with a plank of mass M in vertical plane as shown in the figure. Plank is slightly displaced from equilibrium position in vertical plane such that plank always remain in horizontal position. The spring constant of springs are shown in the figure.

If angular frequency of oscillation of plank is $\sqrt{\frac{8\lambda k}{11m}}$, then find the value of λ .



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

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Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. EMF of the cell: $\text{Cd(s)} | \text{CdCl}_2 \cdot 5\text{H}_2\text{O (saturated solution)} | \text{AgCl(s)} | \text{Ag(s)}$ is +0.70 V at 0°C and +0.60 V at 50°C. If ΔH° and ΔS° are temperature independent, then the correct information(s) regarding the cell reaction is/are

(A) $\Delta G^\circ = -115.8 \text{ kJ at } 50^\circ\text{C}$

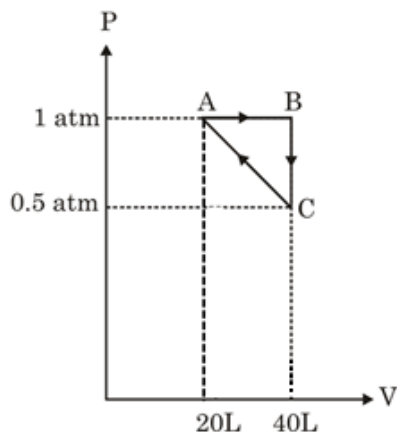
(B) $\Delta G^\circ = -135.1 \text{ kJ at } 0^\circ\text{C}$

(C) $\Delta S^\circ = -386 \text{ J/K}$

(D) $\Delta H^\circ = -221.178 \text{ kJ}$

Space for Rough Work

2. Select the correct option(s) about P–V plot given below for an ideal gas undergoing cyclic process :



[Given : $\ln 2 = 0.7$, $1 \text{ L} \cdot \text{atm} = 100 \text{ J}$]

- (A) The entropy change for the overall process is zero
 (B) $\Delta H > \Delta E$ for overall process
 (C) $W_{\text{net}} = q_{\text{net}}$ (for overall cycle)
 (D) Total work done = – 600 J

3. Among the following choose the correct options :

1. $[\text{Fe}(\text{CN})_6]^{3-}$ (X) and $[\text{FeF}_6]^{3-}$ (Y)
2. $[\text{Fe}(\text{CN})_6]^{3-}$ (S) and $[\text{Fe}(\text{CN})_6]^{4-}$ (T)
3. $[\text{M}(\text{CN})_6]^{-4}$ (P) and $[\text{M}(\text{CN})_4]^{-2}$ (Q)

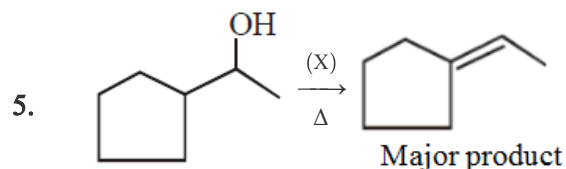
(If valence shell electronic configuration of metal 'M' is $ns^2 (n-1)d^8$)

- (A) 1 - X, 2 - T, 3 - Q (Stability)
 (B) 1 - X, 2 - S, 3 - Q (Crystal field splitting energy)
 (C) 1 - X, 2 - S, 3 - Q (Stability)
 (D) 1 - Y, 2 - T, 3 - Q (Crystal field splitting energy)

Space for Rough Work

4. When crystals of potassium chloride is heated with MnO_2 and conc. H_2SO_4 then

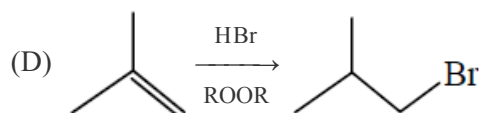
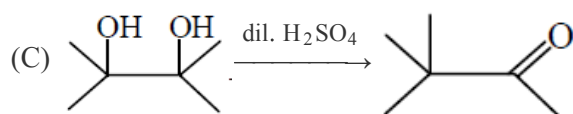
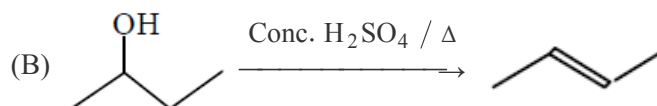
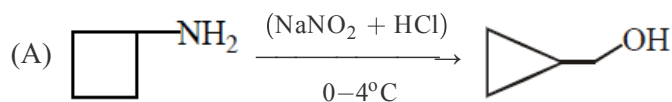
- (A) Cl_2 is produced
- (B) Its redox change
- (C) Resulting mixture after dilution by water gives buff colour ppt. with H_2S
- (D) Resulting mixture after dilution by water gives ppt. with excess. KOH



X can be -

- (A) Conc. H_2SO_4
- (B) Al_2O_3
- (C) P_2O_5
- (D) Conc. H_3PO_4

6. The reaction among following which are correctly match with their Major product



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
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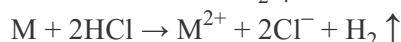
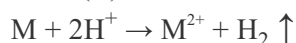
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Negative Marks : -1 In all other cases.

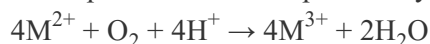
Paragraph for Questions 7 and 8

Metal (**M**) is a white, crystalline metal and is not appreciably ductile or malleable. It has very high melting point. The metal is soluble in dilute or concentrated hydrochloric acid. If air is excluded, Metal(II) ions are formed :

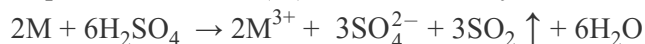


Metal(II) ions are rather unstable as they are strong reducing agent, they decompose even in acidic solution with the formation of hydrogen and M^{+3} .

In the presence of atmospheric oxygen, metal (**M**) gets partly or wholly oxidized to the trivalent state :



Dilute sulphuric acid attacks metal (**M**) slowly, with the formation of hydrogen. In hot, concentrated sulphuric acid metal (**M**) dissolves readily, to form metal(III) ions and sulphur dioxide:



Highly concentrated nitric acid render metal (**M**) passive.

7. Colour of M^{+2} ion & M^{+3} ion respectively when obtain by dissolving MO & M_2O_3 in dil. HCl.
 (A) Blue, Green (B) Green, Blue (C) Blue, Yellow (D) Violet, Blue
8. Aqueous solution of $MCl_3 \cdot 6H_2O$ when reacts with Na_2O_2 it produce coloured solution(s). Coloured solution(s) does not produce any ppt with ($BaCl_2 + \text{dil. HCl}$), but yellow ppt is produced with ($BaCl_2 + CH_3COOH_{aq.}$). Select incorrect about $MCl_3 \cdot 6H_2O$.
 (A) Produce violet-pink complex with excess NH_4OH (aq.)
 (B) Produce light green solution with excess $NaOH_{aq.}$
 (C) Produce M_2S_3 when react with aq. solution of $(NH_4)_2S$
 (D) It is paramagnetic

Space for Rough Work

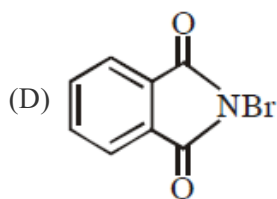
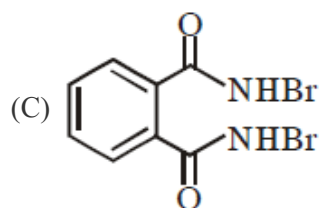
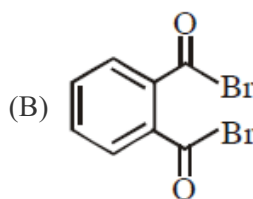
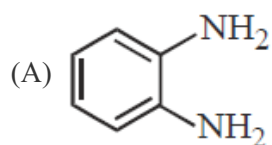
Paragraph for Questions 9 and 10

Treatment of compound O with $\text{KMnO}_4 / \text{H}^+$ gave P, which on heating with ammonia gave Q. The compound Q on treatment with $\text{Br}_2 / \text{NaOH}$ produced R. On strong heating, Q gave S, which on further treatment with ethyl 2-bromopropionate in the presence of KOH followed by acidification, gave a compound T.



(O)

9. The compound R is :



10. The compound T is :

(A) Glycine

(B) Alanine

(C) Valine

(D) Serine

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

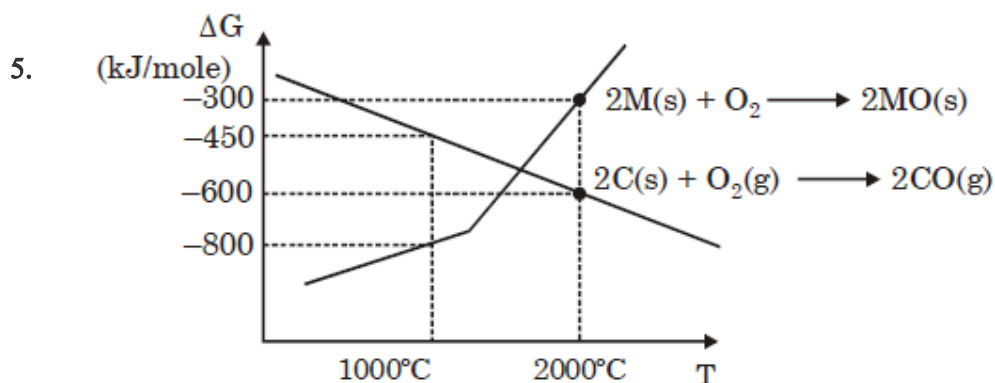
- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

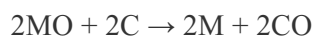
Zero Marks : 0 In all other cases.

-
1. 30 moles of carbon were burnt with 'X' moles of O_2 . If product mixture has 60% by mole CO_2 , 15% by mole CO and 25% by mole O_2 , the value of 'X' is ?
 2. During a reversible adiabatic process, the pressure of an ideal gas is found to be proportional to the cube of its absolute temperature. Now in a different process 1 mole of this same gas is isochorically heated from 1000 K to 2000 K then calculate heat gained by the gas. (in kcal) .
 3. The boiling point of pure benzene is 353.23 K. When 1.80 g of a non-volatile solute was dissolved in 90 g of benzene, the boiling point is raised to 354.11K. Calculate the molar mass (g/mol) of the solute. K_b for benzene is $2.53 \text{ K kg mol}^{-1}$.
 4. If 800 ml of aqueous NaBr solution is electrolysed with a current of 0.193A, how many seconds would it take for the pH of solution to rise to a value of 9 ?
-

Space for Rough Work



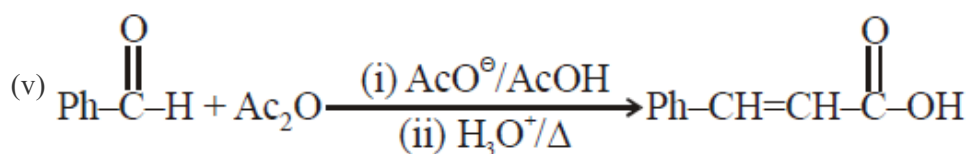
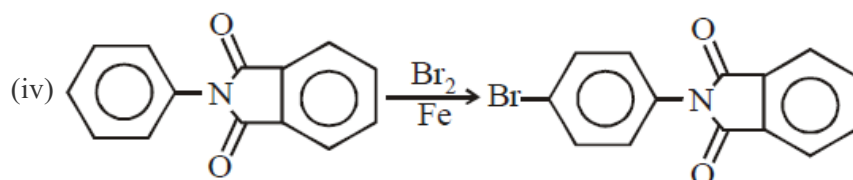
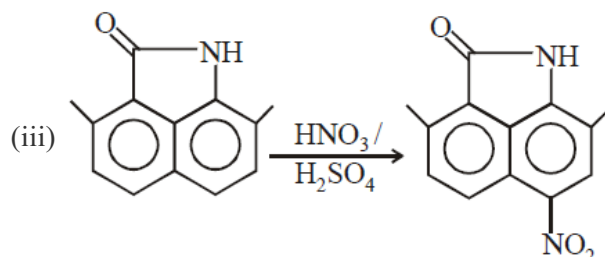
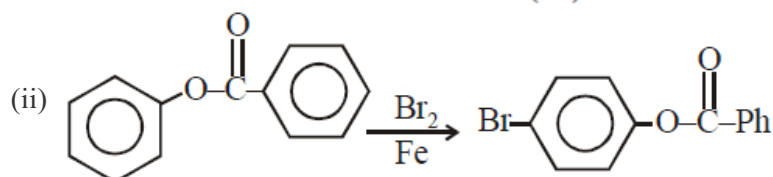
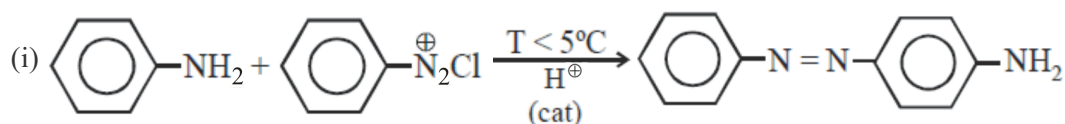
Find the magnitude of the free energy change (in kJ/mole) at the temperature (2000 $^{\circ}\text{C}$) where carbon can reduce the metal oxide to produce CO as follows:



6. Number of reactions which produce O_2 .
- Thermal decomposition of KClO_3
 - Thermal decomposition of PbO_2
 - Thermal decomposition of HgO
 - Thermal decomposition of NH_4NO_2
 - Electrolysis of water
 - Thermal decomposition of KO_2
 - Reaction of $\text{Na}_2\text{O}_2 + \text{CO}_2$

Space for Rough Work

7. Identify reactions which are correctly match with their product ?



8. An alkene with molecular formula C_6H_{12} [A], On ozonolysis produces only one product (B) which gives 2, 4-DNP test positive but Iodoform test negative. What will be the number of a H atoms present in that alkene (A).

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. $z = x + iy$, $x, y \in \mathbb{R}$, $|z| = 1$ and $f(z) = |1 + z| + |1 - z + z^2|$, then
 (A) Maximum value of $f(z)$ is $\frac{13}{4}$ (B) Minimum value of $f(z)$ is $(e - 1)$
 (C) Maximum value of $f(z)$ is 3 (D) Minimum value of $f(z)$ is $\sqrt{3}$
2. Let $f, g : \mathbb{R} \rightarrow [1, \infty)$ are two differentiable function on the real line satisfying the differential equation $f'(f^5 + g^2f) + g'(f^2g + g^3) = 0$, then
 (A) f is bounded but g is not bounded (B) g is bounded but f is unbounded
 (C) both f and g are bounded (D) both f and g are unbounded
3. Let $I_1 = \int_0^{\frac{\pi}{2024}} \sin(2\pi \cos^2(1012x)) dx$, $I_2 = \int_0^{\frac{\pi}{2024}} \sin(\pi \cos^2(1012x)) dx$, $I_3 = \int_0^{\frac{\pi}{2024}} \sin(\pi \sin^2(506x)) dx$, then
 (A) $I_1 = I_2$ (B) $I_2 = I_3$ (C) $I_3 \neq I_1$ (D) $I_1 \neq I_2$

Space for Rough Work

4. Let position vectors of the point A, B, C are $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\vec{c} = 5\hat{i} + \hat{j} - \hat{k}$ respectively, then area of region formed by point R ($\vec{r}$), such that $\vec{r} \cdot \vec{a} = 5$ and $|\vec{r} - \vec{b}| + |\vec{r} - \vec{c}| = 4$
- (A) $\frac{\pi}{\sqrt{2}}$ (B) $\sqrt{2}\pi$ (C) $\frac{\pi}{2\sqrt{2}}$ (D) $2\sqrt{2}\pi$
5. P and Q are two $n \times n$ matrices such that $|P| \neq 0$, $P + Q = (PQ)^2$ and $QPQ = P + I$, then which of the following options is/are correct ?
- (A) $P^{-1} = P^4 - I$ (B) $Q^5 - P^5 = P + Q$ (C) $P^9 = P^4 + P + I$ (D) $P^2Q^2 = QP^2Q$
6. Which of the following option's is/are not always correct ?
- (A) If both functions $y = f(x)$ and $y = g(x)$ are differentiable on the interval (a, b) and intersect some where on (a, b) then the function $\max\{f(x), g(x)\}$ is not differentiable at the points where $f(x) = g(x)$
- (B) If functions $y = f(x)$ and $y = g(x)$ are differentiable and $f(x) > g(x)$ on the interval (a, b) then $f'(x) > g'(x)$ on (a, b)
- (C) If a function $y = f(x)$ is continuous and increasing at the point $x = a$, then there is always a neighbourhood $(a - \delta, a + \delta)$, $\delta > 0$ where the function is also increasing
- (D) A function always has a local maximum between any two local minima

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

If the confocals through point (x_1, y_1) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are $\frac{x^2}{a^2 + \lambda_1} + \frac{y^2}{b^2 + \lambda_1} = 1$ and

$\frac{x^2}{a^2 + \lambda_2} + \frac{y^2}{b^2 + \lambda_2} = 1$, then

7. If point (x_1, y_1) lies inside of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then
- (A) $\lambda_1 \lambda_2 < 0$ (B) $\lambda_1 \lambda_2 > 0$
 (C) $\lambda_1 \lambda_2 = 0$ (D) λ_1 & λ_2 are not real
8. If point (x_1, y_1) lies outside of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and inside of its director circle then
- (A) $\lambda_1 + \lambda_2 > 0$ (B) $\lambda_1 + \lambda_2 < 0$
 (C) $\lambda_1 + \lambda_2 = 0$ (D) $\lambda_1 + \lambda_2$ can be positive or negative

Space for Rough Work

Paragraph for Questions 9 and 10

Consider all the strings of n -digits which can be formed using the numbers from the set $\{1, 2, 3, 4, 5\}$. If P_n denotes the number of strings containing even number of 4's and Q_n denotes the strings containing odd number of 5's, then

9. P_n is equal to

(A) $\frac{5^n - 3^n}{2}$

(B) $\frac{5^n + 3^n}{2}$

(C) $\frac{4^n + 2^n}{2}$

(D) $\frac{4^n - 2^n}{2}$

10. Q_n is equal to

(A) $\frac{5^n - 3^n}{2}$

(B) $\frac{5^n + 3^n}{2}$

(C) $\frac{4^n + 2^n}{2}$

(D) $\frac{4^n - 2^n}{2}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let number of ways to arrange a, a, a, a, b, b, b, b, c, c, d, d in a line so that there must be at least one b's in between any two a's is 'k' then sum of digits of 'k' is
2. Let $f(x) = 4 \left(\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) \right)^2 + 8 \left(\tan^{-1} 2 - \tan^{-1} \frac{1}{3} \right) \cos^{-1} x + \frac{\pi^2}{4}$. If minimum value of $f(x)$ is k, then value of $\frac{\pi^2}{k}$ is
3. Let A and B be $n \times n$ matrices with real entries satisfying $\text{tr}(ABB'A' + BAA'B') = \text{tr}(ABA'B' + BAB'A')$ then value of $|AB - BA|$ is
4. Three alternate numbers a, b, c are removed from 'n' numbers listed 1, 3, 5, 7, 9 ... if the arithmetic mean of the remaining number is $\frac{105}{2}$, then sum of removed numbers is

Space for Rough Work

5. If $f(x) = 2x^3 - 3x^2 + 1$, then number of distinct real solutions of the equation $f(f(x)) = 0$ is
6. Consider the parabola $y = ax - bx^2$, if the least positive value of 'a' for which there exist $\alpha, \beta \in \mathbb{R} - \{0\}$, such that both the point (α, β) and (β, α) lies on the given parabola is 'k', then $[k]$ is (where $[.]$ denotes greatest integer)
7. Let $x_1, x_2 \dots x_{10}$ be real numbers in the interval $\left[0, \frac{\pi}{2}\right]$, such that $\sin^2 x_1 + \sin^2 x_2 + \dots + \sin^2 x_{10} = 1$, if $\frac{\cos x_1 + \cos x_2 + \dots + \cos x_{10}}{\sin x_1 + \sin x_2 + \dots + \sin x_{10}} \geq \alpha$, then max. value of α is
8. The number of four digit numbers without repetition that can be formed from the digits 0, 1, 2, 3, 4, 5, 6, 7, so that each number contains digit 1 is
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work